

HAMEENLINNA KATINEN, Finland

WMO: 027540

Lat: 60.9992N

Lon: 24.4917E

Elev: 85

StdP: 100.31

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-19.7	-25.4	0.4	-22.6	-22.1	0.5	-19.4	10.0	1.3	9.1	1.4	1.1	100	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	27.6	18.8	25.6	17.7	23.8	16.9	20.5	25.0	19.2	23.8	18.0	22.2	3.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	13.9	22.7	17.5	12.7	21.3	16.2	11.6	19.9	59.2	24.9	55.0	23.5	51.0	22.3	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	8.0	7.1	-25.6	29.3	3.5	2.4	-28.1	31.1	-30.1	32.5	-32.1	33.8	-34.6	35.6		
			-24.9	21.2	4.5	1.8	-28.1	22.5	-30.7	23.5	-33.3	24.5	-36.5	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-7.1	-4.8	-1.7	4.4	11.0	14.4	17.9	16.0	11.7	5.0	1.6	-2.6
	DBStd	9.32	6.62	6.20	4.54	3.55	4.24	3.49	3.26	2.96	3.05	3.55	4.11	5.57
	HDD10.0	2342	529	414	362	173	39	4	0	1	18	158	252	391
	HDD18.3	4730	788	648	621	419	229	125	49	86	201	413	502	649
	CDD10.0	712	0	0	0	4	69	138	244	185	68	3	0	0
	CDD18.3	57	0	0	0	0	2	8	35	12	0	0	0	0
	CDH23.3	485	0	0	0	0	40	87	279	79	0	0	0	0
	CDH26.7	88	0	0	0	0	3	12	58	15	0	0	0	0
Wind	WSAvg	3.4	3.2	3.2	3.6	3.4	3.5	3.6	3.3	3.2	3.2	3.5	3.7	3.8
Precipitation	PrecAvg	637	53	40	34	33	37	64	72	76	52	65	59	53
	PrecMax	774	102	84	68	69	72	113	187	162	147	167	141	118
	PrecMin	533	8	2	5	5	10	26	3	14	10	15	5	5
	PrecStd	73	22	22	15	17	18	24	41	33	33	28	25	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	5.6	10.3	20.1	27.1	28.4	30.3	28.7	21.3	14.4	10.2	8.1
		MCWB	3.4	3.0	5.3	11.0	15.6	17.8	19.9	19.6	16.1	11.9	9.1	7.5
	2%	DB	3.3	3.7	7.3	16.6	24.3	25.9	28.3	25.3	19.4	12.5	9.0	5.7
		MCWB	2.6	2.5	3.6	8.9	14.7	16.5	19.9	18.1	15.2	10.8	8.3	4.7
	5%	DB	2.1	2.6	5.4	13.9	21.9	23.7	26.5	23.3	17.8	11.5	7.6	4.5
		MCWB	1.6	1.8	2.8	7.7	13.8	15.7	18.8	16.9	14.0	10.3	6.8	3.8
	10%	DB	1.0	1.7	3.8	11.2	19.5	21.6	24.5	21.6	16.5	10.4	6.5	3.3
		MCWB	0.5	1.2	1.9	5.9	12.3	14.8	18.3	16.3	13.7	9.2	5.8	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.5	5.9	11.5	17.6	20.2	22.0	21.6	16.7	12.7	9.5	7.5
		MCDB	3.9	4.8	8.5	18.0	23.0	24.3	27.2	26.1	19.5	14.0	10.1	8.1
	2%	WB	2.7	2.7	4.2	9.8	16.0	18.1	21.0	19.4	15.9	11.5	8.1	5.0
		MCDB	3.2	3.5	6.5	15.7	21.6	23.9	25.6	23.3	18.4	12.1	8.8	5.4
	5%	WB	1.8	2.0	3.2	8.1	14.7	16.7	20.1	18.1	15.0	10.5	7.0	3.9
		MCDB	2.2	2.6	4.9	12.6	20.2	22.1	24.8	21.9	17.0	11.3	7.6	4.5
	10%	WB	0.7	1.2	2.2	6.7	13.1	15.5	18.9	17.0	14.1	9.2	6.0	2.8
		MCDB	1.0	1.7	3.5	10.5	18.5	20.1	23.6	20.4	15.8	10.1	6.4	3.3
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.2	7.3	9.2	10.3	9.1	8.7	8.5	7.1	5.7	3.6	4.0
		MCDBR	3.7	4.7	8.4	14.2	14.6	12.9	11.6	11.5	9.4	6.8	4.2	4.2
		MCWBR	3.6	4.0	5.5	7.7	6.7	5.5	4.6	5.2	5.4	5.1	4.0	4.1
	5% WB	MCDBR	3.5	4.2	7.4	12.2	12.2	11.1	10.0	9.6	7.9	5.7	4.1	4.3
		MCWBR	3.5	3.7	5.3	7.0	6.3	5.3	4.6	4.9	5.0	5.0	3.9	4.2
Clear-Sky Solar Irradiance	taub	0.259	0.299	0.326	0.357	0.355	0.351	0.375	0.363	0.338	0.324	0.284	0.371	
	taud	2.095	2.165	2.245	2.368	2.433	2.464	2.417	2.443	2.509	2.405	2.194	2.619	
	Ebn at Noon	461	669	788	837	870	878	844	824	786	653	431	315	
	Edh at Noon	40	71	92	99	100	99	102	92	72	57	37	26	
All-Sky Solar Radiation	RadAvg	0.24	0.80	2.28	3.76	5.07	5.29	5.12	3.96	2.36	1.02	0.28	0.13	
	RadStd	0.04	0.18	0.29	0.39	0.56	0.42	0.50	0.44	0.31	0.18	0.05	0.02	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (61 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	-145	N/A	N/A	N/A	N/A

Nomenclature: See separate page